The same practice budget, more relevant evidence

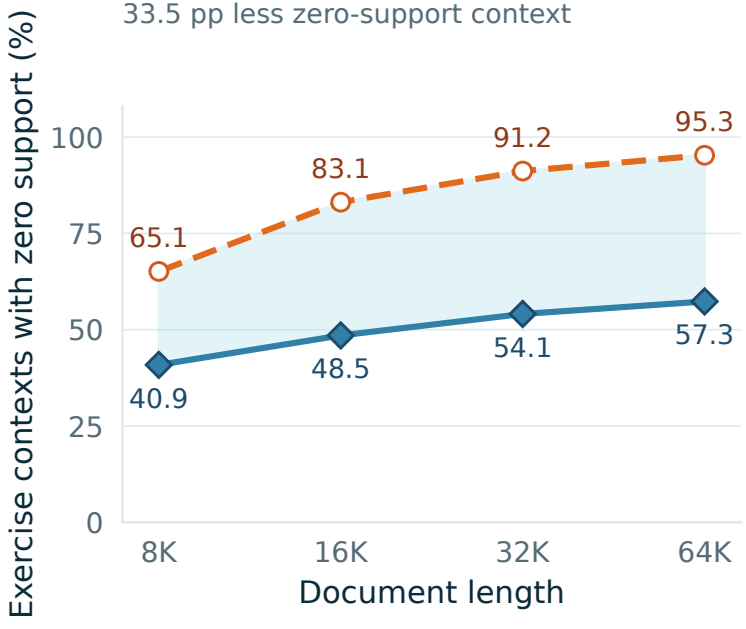
BABILong · Qwen3-4B · 1,000 document–query pairs · 26,451 matched synthetic questions per method

Vanilla Cartridges data — Query-specific data (ours)

a DILUTION

83.7 → 50.2%

33.5 pp less zero-support context

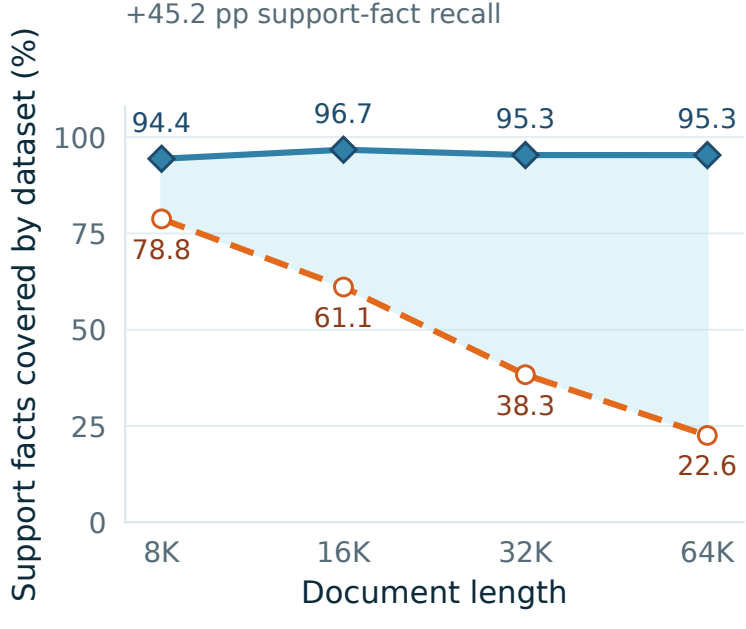


1,000 paired cases · 250 per length

b SUPPORT COVERAGE

50.2 → 95.4%

+45.2 pp support-fact recall

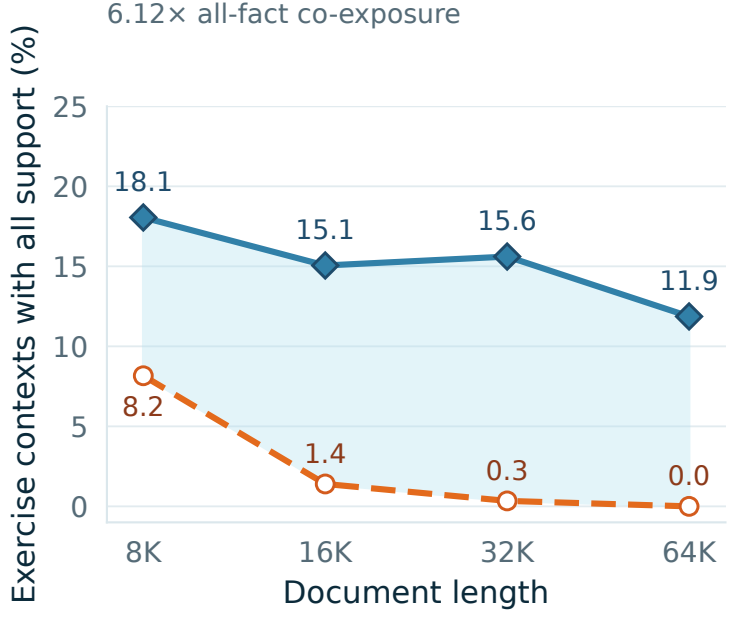


994 paired cases · 6 missing excluded

c FRAGMENTATION

2.5 → 15.1%

6.12× all-fact co-exposure



352 multi-fact cases · 88 per length

Equal-weight case means. Shading marks the gap between methods. Coverage uses post-generation supporting-fact annotations.